

Building a gm/ID Lookup Table in Cadence Virtuoso Using 180 nm CMOS Technology

1. Aim

To develop a gm/ID-based lookup table using 180 nm CMOS technology for calculating the required transistor parameters and dimensions.

2. Requirement

Technology: 180 nm CMOS

Channel length L: 180 nm

VDS parametric sweep: 0 to 1.8 with six steps

Width W parametric sweep: 400n to 2u with six steps

DC analysis sweep step: 10 mV

Plot gm/ID versus ID/W

Plot gm/ID versus Vov

3. Theory

The transconductance gm shows how effectively the gate voltage controls the drain current ID.

$$g_m = \frac{I_D}{V_{GS}}$$

Overdrive voltage is given by:

$$V_{OV} = V_{GS} - V_{TH}$$

Lookup curves of gm/ID versus ID/W and gm/ID versus Vov can be used to select the transistor operating point and calculate the required transistor width.

4. Experimental Setup

Experiment was performed using Cadence Virtuoso with 180 nm CMOS technology. MOS transistor test circuit was designed to generate the gm/ID lookup table.

The schematic used for the simulation is shown below.

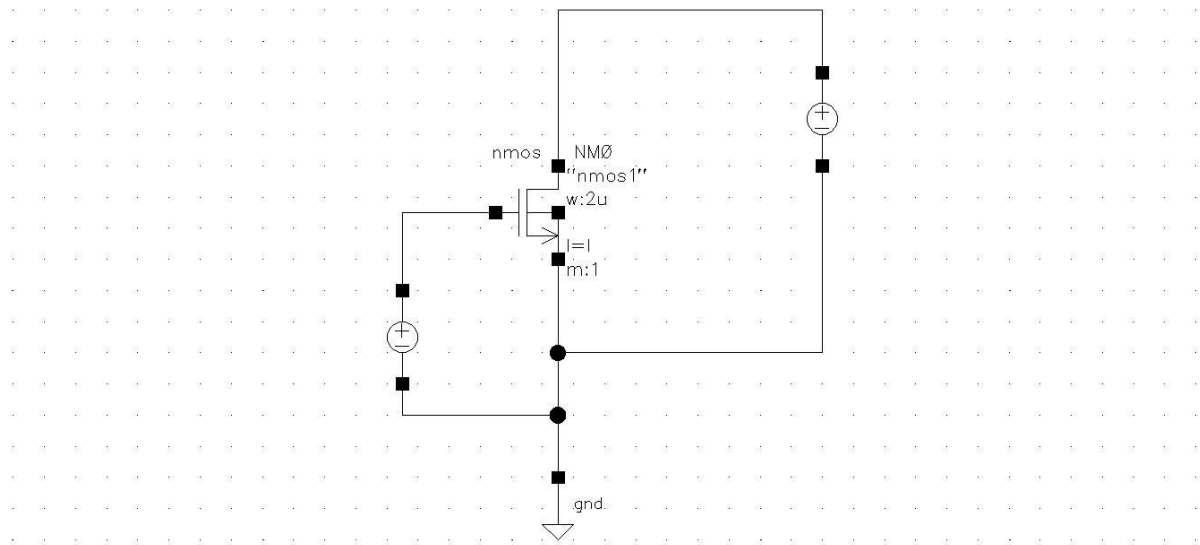


Figure 1: MOS transistor test circuit used for generating the gm/ID lookup table.

5. ADE L Setup & Parametric sweep

Simulation performed using Cadence ADE L with 180 nm CMOS technology. DC analysis was carried out by sweeping VGS from 0 V to 1.8 V with a step size of 10 mV. In the parametric analysis, VDS was swept from 0 V to 1.8 V, while the transistor width W was swept from 400 nm to 2 um with six sweep points each.

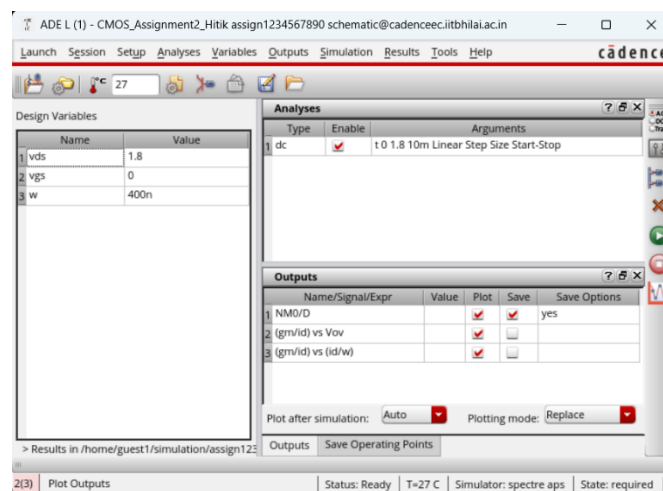


Figure 2: ADE L setup showing the design variables, DC analysis, and output expressions.

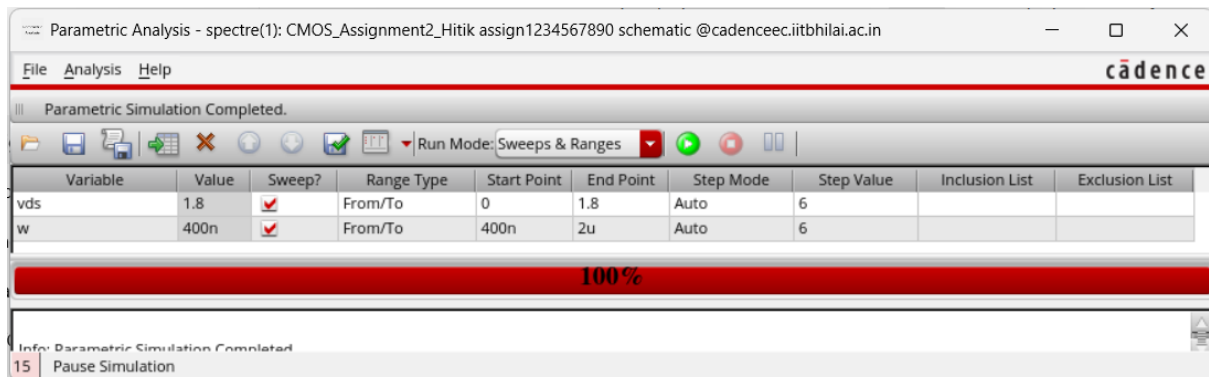


Figure 3: Parametric sweep setup for VDS and transistor width W.

6. Results and Discussion

6.1 gm/ID versus Vov

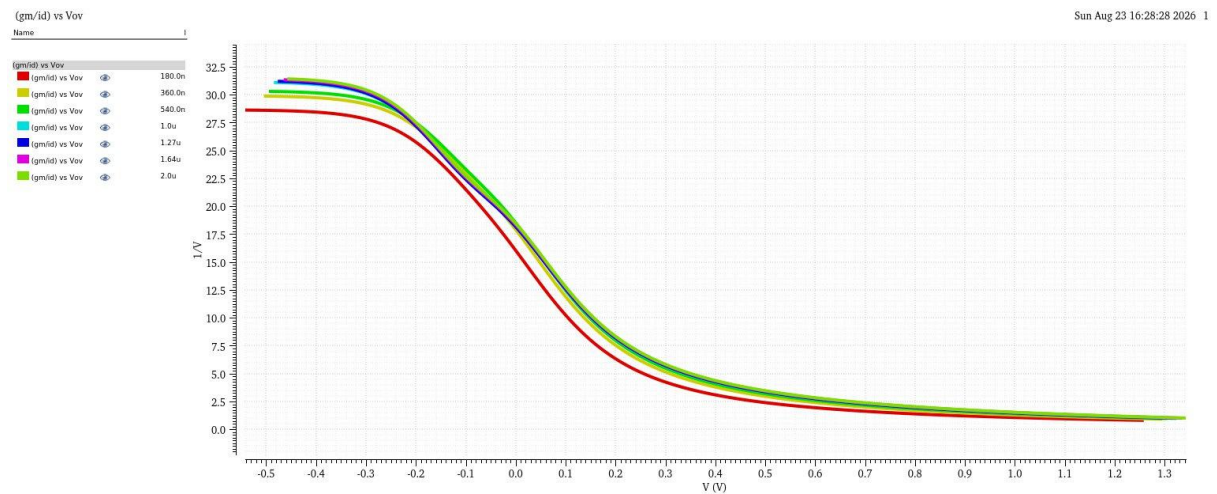


Figure 4: Variation of gm/ID wrt overdrive voltage Vov. As Vov increases, gm/ID decrease, indicating the transition from weak inversion towards strong inversion.

6.2 gm/ID versus ID/W

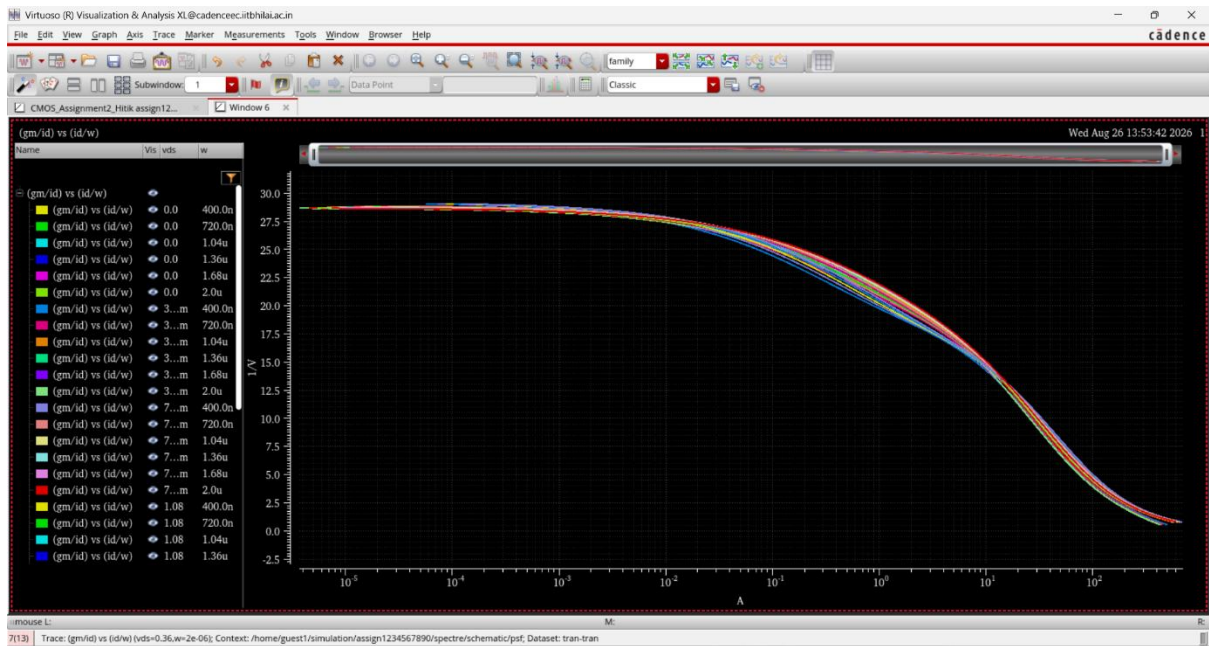


Figure 5: gm/ID plotted against normalized current ID/W for different transistor widths or VDS conditions. The graph can be used to select the required operating point and determine the transistor width.

7. Look-up Table

<https://docs.google.com/spreadsheets/d/e/2PACX-1vQK8Vc3YbnPpPLrGe-xJAn29xA0WQNr90yUeMUhGhUZuUwm7TUK6bIQRME8qNIwMUuTRPSSBIFNC9eS/pu b?output=xlsx>

8. Result & Conclusion

The gm/ID lookup curves were successfully generated using Cadence Virtuoso with 180 nm CMOS technology. The channel length was fixed at 180 nm, while VDS and transistor width W were varied using parametric sweeps. The plots of gm/ID versus Vov and gm/ID versus ID/W were obtained using a DC sweep step size of 10 mV. These lookup curves can be used to select a suitable transistor bias point and determine the required transistor width for circuit design.